3.2.1 Let $f \in A_k(\Gamma)$ be automorphic form and $s \in \mathbb{Q} \cup \{\infty\}$ cusp.

Show $\nu_\infty(f) = m$ $\quad f(\tau) = \sum_{n=m}^{\infty} a_n q_h^n$ truncated from left at m

$\nu_s(f) = \nu_\infty(f[\alpha]_k)$

$\mathrm{Im}\,\tau > c \quad q_h = e^{2\pi i \tau/h}$

is independent of choice of α, and well-defined on $X(\Gamma)$.

Pf: i) If $\alpha\infty = \beta\infty = s$ then $\cancel{\exists \gamma \in SL_2(\mathbb{Z}) \quad \alpha = \gamma\beta}$.

$\alpha^{-1}\beta\infty = \infty$

$\Rightarrow \alpha^{-1}\beta = \pm\begin{pmatrix}1 & m\\ 0 & 1\end{pmatrix}$

Since f weakly modular wrt Γ

$$(f[\beta]_k)(\tau) = f[\alpha\begin{pmatrix}1 & m\\ 0 & 1\end{pmatrix}]_k(\tau) = f[\alpha]_k(\tau+m).$$

Thus $g(\tau) := f[\alpha]_k \Rightarrow (f[\beta]_k)(\tau) = g(\tau+m)$.

We want to show $g(\tau) = g(\tau+m)$.

$$g(\tau) = \sum_n a_n e^{2\pi i n\tau/h} \Rightarrow g(\tau+m) = \sum_n a_n e^{2\pi i n\tau/h} e^{2\pi i n m/h}.$$

so the m does not change.

ii) Let $s' = \gamma s$, $\gamma \in \Gamma$. If $\alpha'\infty = s'$

then $\gamma^{-1}\alpha'\infty = s$. $\quad f[\gamma^{-1}\alpha']_k = f[\alpha']_k$

by weakly modular.

3.2.4. Show $\forall k \in \mathbb{Z}$ if $0 \neq f \in \mathcal{A}_k(\Gamma)$

then $\mathcal{A}_k(\Gamma) = \mathbb{C}(X(\Gamma))\, f$.

Pf: Let $g \in \mathcal{A}_k(\Gamma)$. We want to show $h = g/f \in \mathbb{C}(X(\Gamma))$.

i.e. h meromorphic i.e. h modular form.

$$h(\gamma\tau) = \frac{g}{f}(\gamma\tau) = \frac{(c\tau+d)^k g(\tau)}{(c\tau+d)^k f(\tau)} = \frac{g}{f}(\tau).$$

Let $s \in \mathbb{P}^1(\mathbb{Q})$ be a cusp.

$$f[\alpha]_k(\tau) = \sum_{n=n_0}^{\infty} a_n q_h^n, \quad g[\alpha]_k(\tau) = \sum_{n=m_0}^{\infty} b_n q_h^n \quad h = h_s.$$

Then $h[\alpha]_0(\tau) = \dfrac{g[\alpha]_k(\tau)}{f[\alpha]_k(\tau)}$ meromorphic since $f \neq 0$.

We give interpretation $\mathbb{C}(X(\Gamma)) \to \mathcal{A}_k(\Gamma)$

$$h \mapsto hf$$

Thus $\mathcal{A}_k(\Gamma)$ is $\mathbb{C}(X(\Gamma))$-vector space $g/f \leftarrow\!\shortmid g$

of dim 1.

3.2.5. Let $\alpha = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix}$, $\Gamma = \alpha^{-1}\Gamma_1(4)\alpha$. Show $\Gamma_\infty = \left\langle -\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right\rangle$.

Pf: $\begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix} \Big| \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \Big| \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Big| \underbrace{\begin{pmatrix} -1 & 1 \\ -2 & 1 \end{pmatrix}}_{= \alpha^{-1}}$.

$\forall \gamma \in \Gamma_1(4)$ $\quad \gamma \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} (4)$.

Thus $\alpha^{-1}\gamma\alpha \equiv \begin{pmatrix} -1 & * \\ -2-2b+1 & * \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} (4)$.

Since $\alpha^{-1}\gamma\alpha \in SL_2(\mathbb{Z})$, $\alpha^{-1}\gamma\alpha \equiv \pm\begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$

so $\left\langle \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \right\rangle \subset \Gamma_\infty$ for some b.

Let $\gamma = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \in \Gamma_1(4)$ then $\alpha^{-1}\gamma\alpha = \begin{pmatrix} 3 & 3 \\ 0 & 3 \end{pmatrix} \equiv -\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} (4)$

Thus $\left\langle -\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right\rangle \subset \Gamma_\infty$. as there are no others, this is Γ_∞

3.2.6 Verify $g[S]_{24}(\tau) = N^{-12} g(\tau/N)$

$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

$$= N^{-12} q_N^{N+1} \prod_{n=1}^{\infty} (1-q_N^n)^{24} (1-q_N^{Nn})^{24}.$$

$$g(\tau) = \varphi_k(\tau) = \eta(\tau)^{k(N+1)} \eta(N\tau)^{k(N+1)}.$$

Pf:

$$= \left((2\pi)^{-\frac{1}{2}} \Delta(\tau)^{\frac{1}{24}}\right)^{k(N+1)} \left((2\pi)^{-\frac{1}{2}} \Delta(N\tau)^{\frac{1}{24}}\right)^{k(N+1)}$$

$$= (2\pi)^{-k(N+1)} \Delta(\tau)^{\frac{k(N+1)}{24}} \Delta(N\tau)^{\frac{k(N+1)}{24}}$$

$$\overset{k(N+1)=24}{=} (2\pi)^{-24} \Delta(\tau) \Delta(N\tau).$$

~~Recall $\Delta\left(\frac{-1}{\tau}\right) = \tau^{12} \Delta(\tau)$.~~

Thus $g[S]_{24}(\tau) = \tau^{-24} g\left(\frac{-1}{\tau}\right) = \tau^{-24} (2\pi)^{-24} \Delta\left(\frac{-1}{\tau}\right) \Delta\left(\frac{-1}{N\tau}\right)$

$$\boxed{= (2\pi\tau)^{-24} \tau^{12} \Delta(\tau) \tau^{12} \Delta(N\tau)} = (2\pi)^{-24} \Delta(\tau) \Delta(N\tau).$$

$$= \cancel{(2\pi\tau)^{-24} (2\pi)^{12}}$$

Recall $\eta\left(\frac{-1}{\tau}\right) = \sqrt{-i\tau}\, \eta(\tau)$. (prp 1.2.5)

$$g[S]_{24}(\tau) = \tau^{-24} g\left(\frac{-1}{\tau}\right) = \tau^{-24} \eta\left(\frac{-1}{\tau}\right)^{24} \eta\left(\frac{-1}{N\tau}\right)^{24}$$

$$= \tau^{-24} \left(\sqrt{-i\tau}\, \eta(\tau)\right)^{24} \left(\sqrt{-iN\tau}\, \eta(N\tau)\right)$$